Day 23 – Authentication Log Analysis Report

Authentication log analysis is an important activity in cybersecurity that involves examining login records to identify successful and failed authentication attempts. These logs provide information about user activity, timestamps, source IP addresses, and authentication results.

Analyzing authentication logs helps security professionals identify unusual login patterns, detect potential password-guessing attacks, and investigate unauthorized access attempts. This report examines a synthetic authentication dataset to understand login behavior and identify suspicious activity.

Objective

The objective of this task is to develop basic security log analysis skills by examining authentication records. The analysis focuses on identifying successful and failed login attempts, grouping events by username and source IP address, recognizing suspicious patterns, and recommending appropriate security measures.

Tools Used

Microsoft Excel: Used to organize authentication records, filter events, count successful and failed logins, and group login attempts by username and source IP address.

Python: Can be used to automate log analysis, calculate event counts, group records, and identify repeated authentication failures.

Text Editor: Useful for viewing and examining raw authentication log files.

Dataset Description

The dataset contains 20 synthetic authentication records created for practice. Each record includes the following fields:

- **Timestamp:** The date and time when the login attempt occurred.
- **Username:** The account associated with the authentication attempt.
- **Source IP:** The IP address from which the login attempt originated.
- **Status:** Indicates whether the authentication attempt was successful or failed.

The dataset contains seven successful login attempts and thirteen failed login attempts. It includes activity associated with the users alice, bob, admin, charlie, and dana.

The dataset is synthetic and is intended only for educational analysis. It does not represent actual user activity or a confirmed security incident.

	A	B	C	D	E
1	Timestamp	Username	Source IP	Status	
2	23-10-2026 09:00	alice	192.168.1.10	SUCCESS	
3	23-10-2026 09:04	bob	203.0.113.25	FAILURE	
4	23-10-2026 09:04	bob	203.0.113.25	FAILURE	
5	23-10-2026 09:05	bob	203.0.113.25	FAILURE	
6	23-10-2026 09:07	bob	203.0.113.25	FAILURE	
7	23-10-2026 09:12	alice	192.168.1.10	SUCCESS	
8	23-10-2026 09:12	admin	198.51.100.44	FAILURE	
9	23-10-2026 09:12	admin	198.51.100.44	FAILURE	
10	23-10-2026 09:12	admin	198.51.100.44	SUCCESS	
11	23-10-2026 09:18	charlie	192.168.1.35	SUCCESS	
12	23-10-2026 09:24	bob	203.0.113.25	FAILURE	
13	23-10-2026 09:31	dana	192.168.1.50	SUCCESS	
14	23-10-2026 09:35	admin	198.51.100.44	FAILURE	
15	23-10-2026 09:35	admin	198.51.100.44	FAILURE	
16	23-10-2026 09:35	admin	198.51.100.44	FAILURE	
17	23-10-2026 09:42	alice	192.168.1.10	SUCCESS	
18	23-10-2026 09:48	charlie	203.0.113.88	FAILURE	
19	23-10-2026 09:48	charlie	203.0.113.88	FAILURE	
20	23-10-2026 09:48	charlie	203.0.113.88	FAILURE	
21	23-10-2026 09:55	dana	192.168.1.50	SUCCESS	
22					
23					

Analysis Method

The authentication records were organized in Microsoft Excel using separate columns for timestamps, usernames, source IP addresses, and login status.

The total number of login attempts was calculated, and the records were classified according to their authentication status. The analysis identified seven successful attempts and thirteen failed attempts out of twenty total attempts.

The failed login events were then grouped by username and source IP address to identify repeated authentication failures. Particular attention was given to multiple failed attempts occurring within short periods and failed attempts followed by a successful login.

These patterns were treated as indicators requiring further investigation rather than conclusive evidence of malicious activity.

Findings

The analysis produced the following summary:

Metric	Result
Total login attempts	20
Successful logins	7

Failed logins	13
Successful login rate	35%
Failed login rate	65%

Repeated failures associated with bob: The account bob recorded five failed login attempts from the source IP address 203.0.113.25. Several attempts occurred within a short period, which may indicate password guessing or repeated incorrect password entry.

Authentication activity associated with admin: The admin account recorded five failed login attempts and one successful login from 198.51.100.44. Two failed attempts were followed by a successful login, and three additional failures occurred later. This pattern deserves review because administrative accounts require stronger protection.

Repeated failures associated with charlie: The account charlie recorded three consecutive failed login attempts from 203.0.113.88 within approximately thirty seconds. This could indicate repeated password entry errors or a password-guessing attempt.

Successful login activity: The accounts alice and dana each recorded three and two successful login attempts respectively. The account charlie also had one successful login from 192.168.1.35. These events provide context for comparing successful and unsuccessful authentication behavior.

The analysis indicates that failed authentication attempts account for 65% of the records. Although this percentage is high in the synthetic dataset, the records alone cannot establish whether an attack occurred. Additional evidence, such as account ownership, expected login behavior, and other system logs, would be required.

Security Recommendations

- **Monitor failed login attempts:** Establish alerts for repeated failures involving the same account or source IP address.
- **Enable multifactor authentication:** Require additional authentication factors, especially for administrative and privileged accounts.
- **Apply rate limiting:** Limit repeated authentication attempts and consider appropriate temporary lockout policies.
- **Review successful logins after failures:** Investigate successful authentication that follows repeated failed attempts, particularly for privileged accounts.
- **Use strong passwords:** Encourage unique passwords and use secure password storage and authentication practices.
- **Protect authentication logs:** Restrict access to logs and protect them against unauthorized modification or deletion.
- **Correlate security events:** Compare authentication records with system, firewall, and application logs before concluding that an incident has occurred.
- **Investigate suspicious patterns:** Review the context of repeated failures to distinguish legitimate mistakes from potential attacks.

Authentication log analysis helps security professionals understand login behavior and identify patterns that may indicate unauthorized access attempts. In this exercise, twenty synthetic authentication records were analyzed, resulting in seven successful and thirteen failed login attempts.

Repeated failures involving bob, admin, and charlie were identified as patterns requiring further investigation. By combining log analysis with multifactor authentication, monitoring, rate limiting, and secure log management, organizations can improve their ability to detect and respond to authentication-related threats.